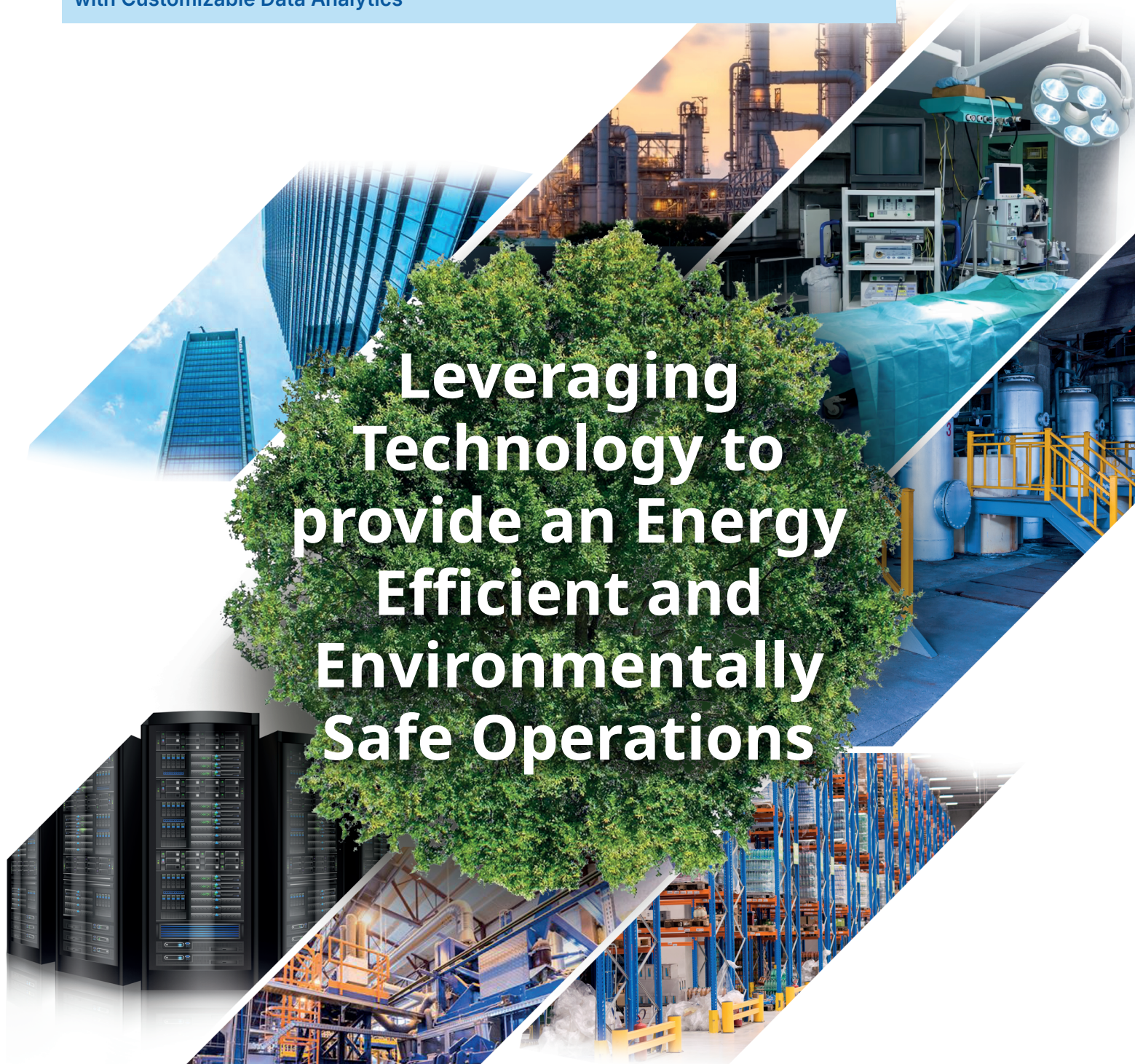


PowerOptimusTM - The Power to Optimize

A real-time data driven Energy Management and Environmental Monitoring System
with Customizable Data Analytics



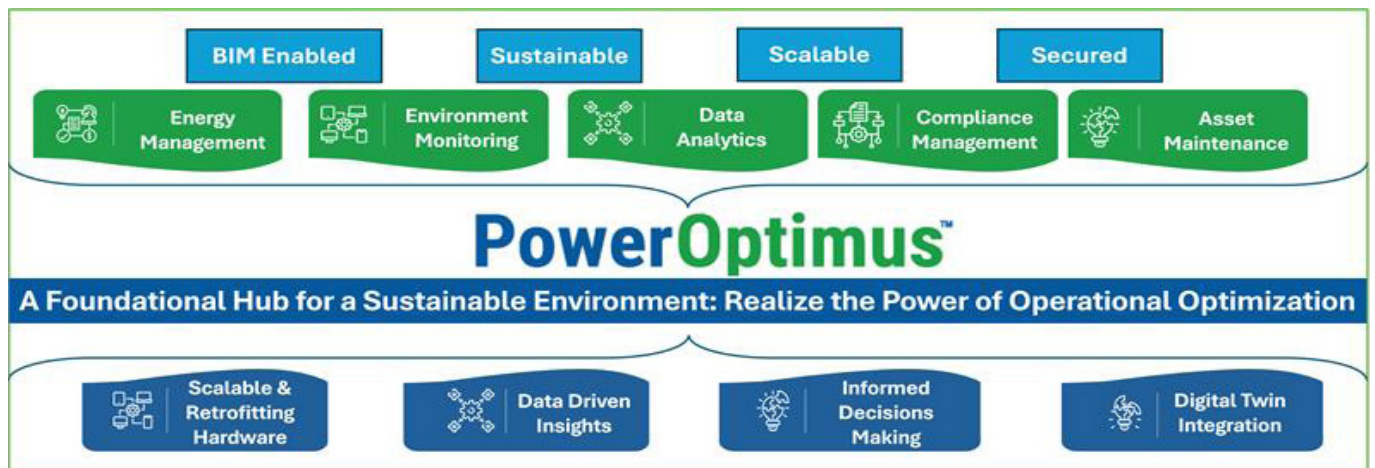
Leveraging
Technology to
provide an Energy
Efficient and
Environmentally
Safe Operations

BIM → EEMS → Digital Twin → Asset Maintenance

CONNECTED | SCALABLE | CUSTOMIZABLE

For

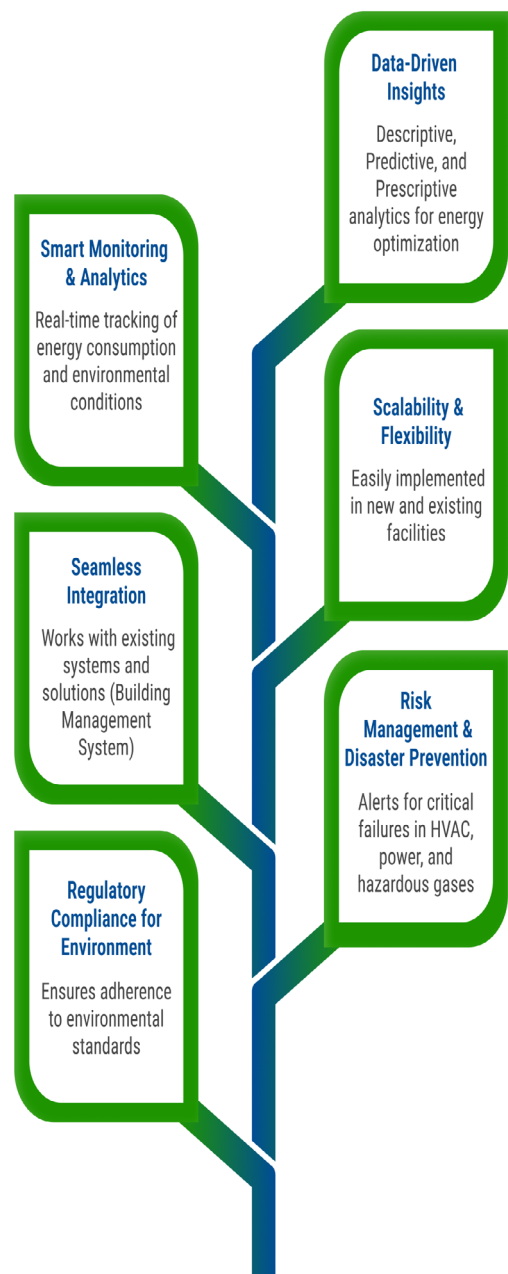
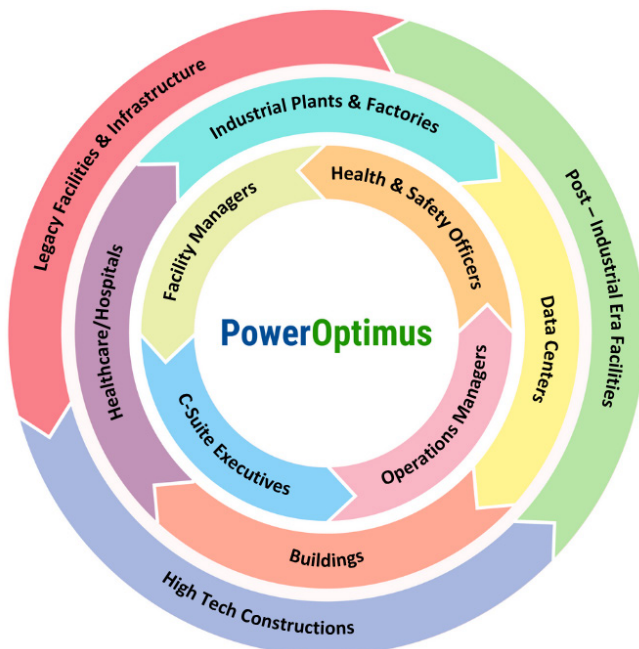
Large Buildings, Industrial Facilities, Data Centers,
Hospitals, Pharma & Chemical Plants, Convention Centers & Malls



PowerOptimus empowers organizations to manage energy consumption and environmental health monitoring with data-driven precision. Leveraging IoT hardware, seamless connectivity, and data-driven analytics, it improves energy efficiency and monitors environmental health

-  Supports compliance with standards such as ISO 50001 (Energy Management), ISO 14001 (Environmental Management), and aligns with ESG objectives mandated by regulatory authorities
-  Facilitates the measurement and notification of deviations when equipment parameters exceed predefined limits
-  Helps optimize Operating Costs
-  Identifies Energy Optimization Opportunities
-  Facilitates consistent monitoring of environmental goals

PowerOptimus seamlessly integrates across a variety of facility types irrespective of their stages - Operational, Under Construction or at Design stage (utilizing mandated BIM methodology). Through uninterrupted data collection across facilities from sensors and meters, it provides the energy optimization and environmental monitoring needs through intuitive dashboards.



Features

Cost-Saving and Efficient

- Potentially reduces energy consumption, cutting operational costs

Beyond BMS

- Fully integrable with the existing BMS, our PowerOptimus adds a layer of insights, enabling smarter decisions, a greater ROI and effective compliance with sustainability standards like ISO 50001 and ISO 14001

BIM Integration- Quick and Easy

- Compatible with existing as-built BIM data that can be used for quick identification and assessment of equipments and systems
- Easy integration with Digital Twin
- Integrate IP-based CCTV cameras to deliver real-time video feeds in a unified cockpit, enabling centralized multi-site surveillance, incident detection, and proactive operational control

What PowerOptimus Monitors:

Environmental Parameters: Temperature, Humidity, Particulate Matter (PM), Greenhouse Gases (CO₂, N₂O, CH₄, ...), Hazardous Gases (CO, NO₂, H₂S, NH₃, Cl₂, O₃, VOC_s...)

Industrial Parameters: Noise, vibration, pressure, water/air flow, gas emissions etc.

Predictive Equipment Maintenance: Early detection of failures in industrial machinery, and critical assets

Energy Sources: Grid power, solar, diesel generators, HVAC systems

Energy Consumers: Industrial machinery, HVAC systems, Elevators, Lighting, Healthcare Equipments etc.

Analytical and Insightful

- **Descriptive Analytics:** Can detect excessive HVAC energy use, machinery wear via noise anomalies, and monitors CO₂, sound, and air quality in hospitals & other similar facilities
- **Predictive Analytics:** Predict potential equipment failure based on monitored parameters and predict energy consumption against maximum demand (Customized Solutions)
- **Prescriptive Analytics:** Recommends optimal HVAC, lighting, and air exchange adjustments based on real-time occupancy and air quality

Retrofittable and Compatible

- No major infrastructure overhauls necessary
- Compatibility with multiple communication protocols ensures integration with existing systems and expansion across facilities and locations

Compliance and Certifications-

- **ISO 14001 – Environmental Management System**

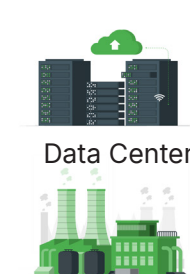
Automates monitoring of greenhouse gases, air quality, hazardous gases and objectives providing real-time alerts and supporting sustainability initiatives through energy optimization strategies

- **ISO 50001 – Energy Management System (EnMS)**

Monitors real-time energy use (grid, solar, diesel), detects inefficiencies, and ensures regulatory compliance



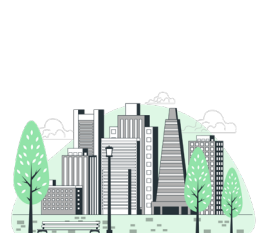
Pharma / Chem



Industrial Plant

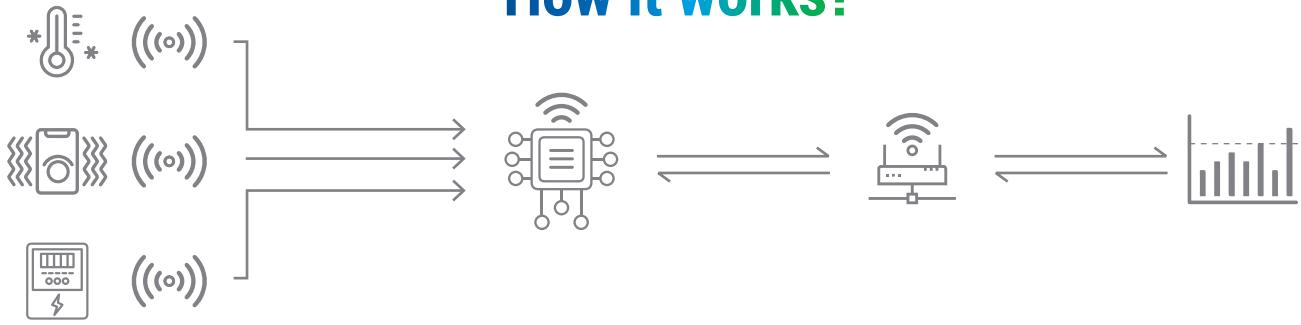


Hospital



Buildings

How it works?



Connect to collect real-time data from energy meters and a variety of sensors [Hazardous Gases, PMI, Noise, Vibrations, Ambient Light, etc.,]

SmartNode

IoT Node supports industry-standard communication protocols to collect data from connected meters and sensors

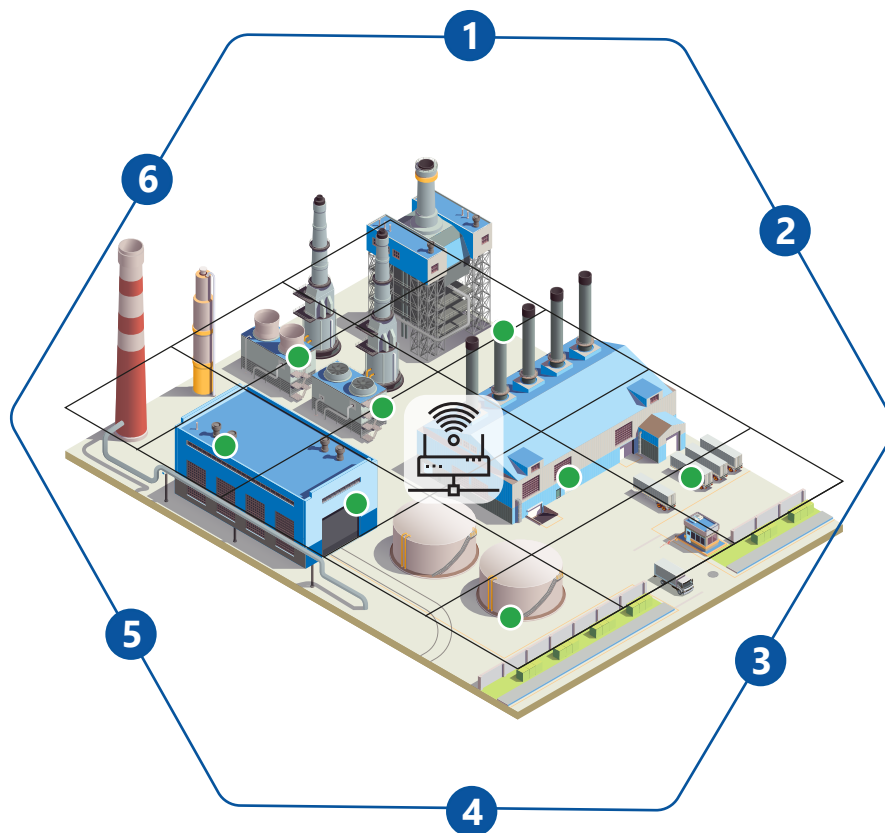
SmartGateway

The collection of IoT Gateway(s) connected to several IoT Nodes establishes Data Fabric to transmit the real-time data

PowerOptimus

Real-time data collected through IoT Gateway(s) is ready for data-driven analytics and visualization

PowerOptimus Implementation Process



1

Facility Assessment

PowerOptimus Experts audit the facility to define objectives, use cases & scope for Energy Monitoring, Optimization and Management, and Environmental aspects

2

Install & Connect PowerOptimus

PowerOptimus connects energy meters and sensors via Smart Node(s) and SmartGateway(s) to establish PowerOptimus Data Fabric

3

Collect Real-time Data

Data Fabric collects and normalizes data at set intervals

4

Customize, Configure & Visualize

Rapidly configure to visualize real-time data & patterns using a variety of charts/ widgets ... live, daily, weekly, trends

5

Data-Driven Analytics

Applies specific use cases providing actionable descriptive and predictive insights

6

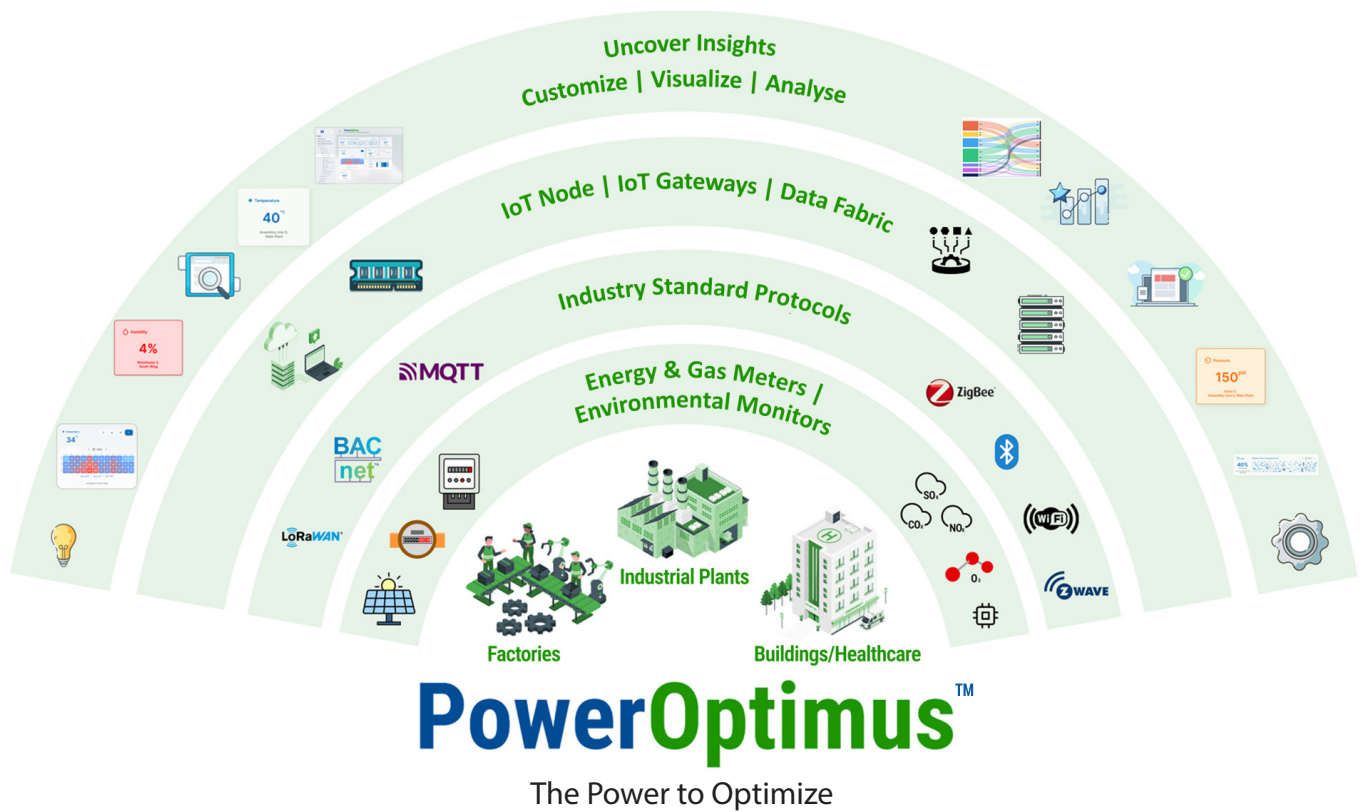
Monitor, Manage, Scale & Elevate

Refine objectives and scope for better coverage to reduce costs incrementally

● PowerOptimus IoT Node

 PowerOptimus IoT Gateway

 Data Fabric



PowerOptimus is built for adaptability and future-readiness, making it ideal for retrofitting into facilities of any age or type with minimal infrastructure disruption. Its support for a wide range of industrial protocols ensures smooth, flexible integration across infrastructures.

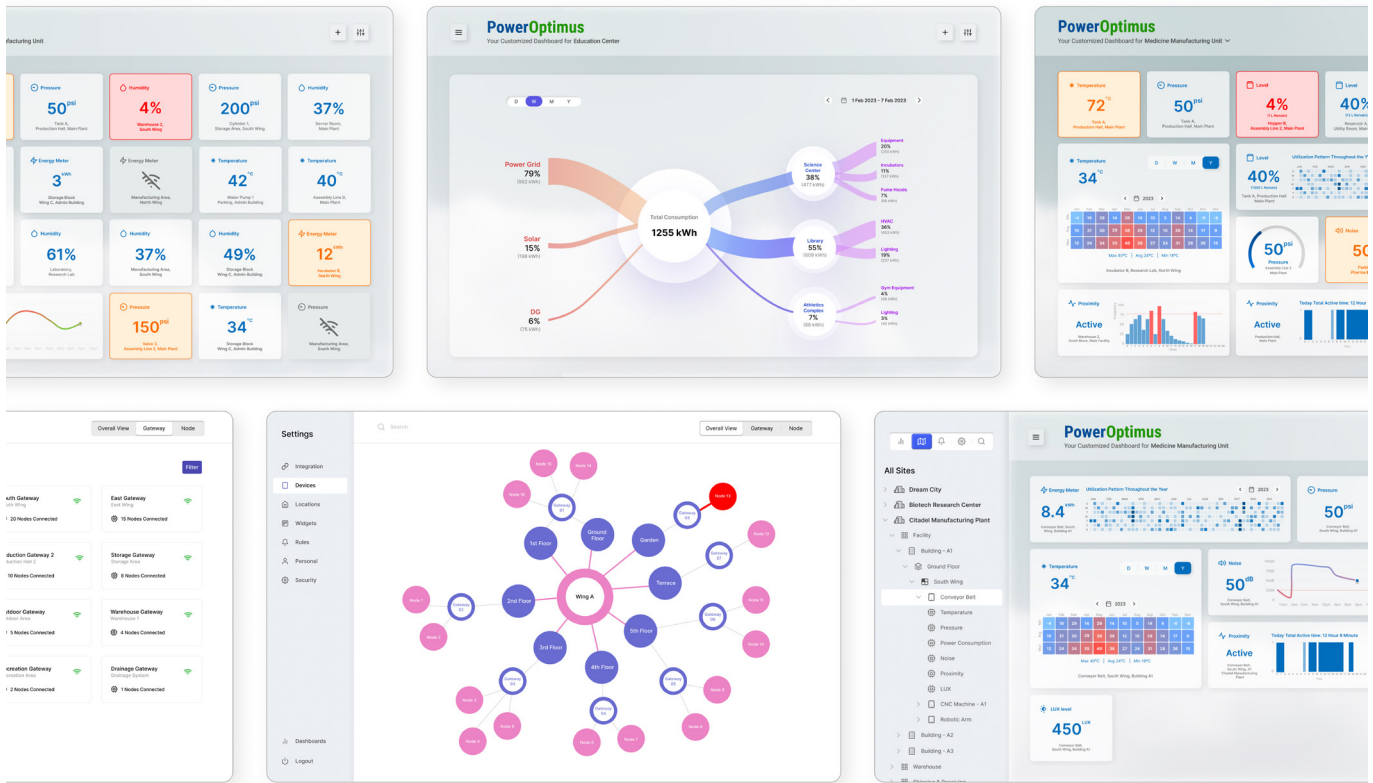
PowerOptimus continuously monitors critical environmental and operational parameters, such as greenhouse and hazardous gases, temperature, humidity, noise, vibrations and other equipment health indicators. Through smart IoT nodes and gateways, it collects, structures and analyzes this data in real-time. Interactive dashboards deliver actionable insights, enabling predictive maintenance, and improved environmental compliance.

With a strong focus on environmental awareness, operational resilience and sustainability, PowerOptimus empowers organizations to operate smarter, safer and greener.



Customizable Intuitive Dashboards

Access real-time data through a modern web-based user interface that offers a range of charts, graphs and tables for easy visualization.



PowerOptimus is your partner in driving sustainability goals, safe and smarter operations. From conducting in-depth on-site assessments to identifying improvement opportunities in environmental performance and asset health, PowerOptimus empowers your facility with data-driven insights and guidance. Its data-powered analytics and predictive maintenance capabilities help optimize operational efficiency while minimizing environmental impact. Through ongoing evaluations and intelligent refinement, PowerOptimus enables continuous improvement, supports compliance and delivers long-term sustainable value across your facility.



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Neil Automation is a wholly owned subsidiary of Neilsoft, that is focused on providing Automation & Sustainability Solutions for Manufacturing Plants, Large Buildings, and Industrial Facilities. The USP of Neil Automation lies in leveraging the latest technologies and evolving cost-effective solutions for the Automation & Sustainability goals of organizations.

Neilsoft, established in 1993, is a global Engineering Services & Solutions company, headquartered in Pune, India addressing clients in Manufacturing, Construction, and Process Industries.

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